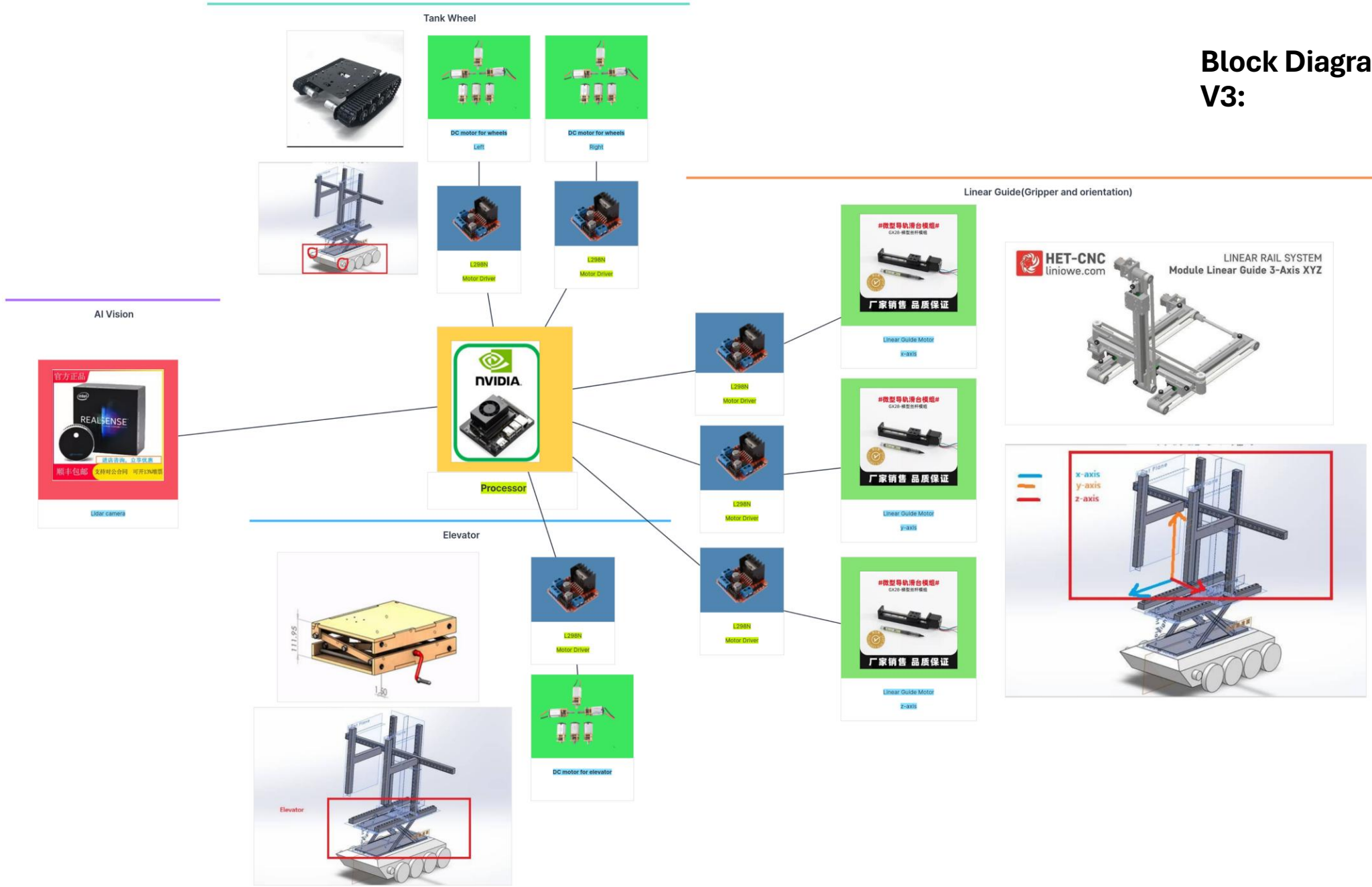
IC382 A04

System Design

Block Diagram V3:



Component List:

1. L298N Quantity: 6	https://world.taobao.com/item/L1phQmhVK1pkSHRxZnRHbVc3K3JMUT09.htm?spm=a21wu.10013406-tw.taglist-content.2.342b63fbjrXeaW 
2. DC motor for wheels + elevator Quantity: 3	https://world.taobao.com/item/WTdrcXZrc0xqWnNjTjQ3blhDeEh6QT09.htm?spm=a21wu.10013406-tw.taglist-content.9.2ee1752b74bDSv4 
3. Linear guide motor Quantity: 3	https://sg.world.taobao.com/item/dzJ5MndLeENlYzFvbnZZdVZEOW5hZz09.htm?spm=a21wu.10013406-sg.taglist-content.2.1dba44a2f55oqE 
4. Lidar Camera Quantity: 1	https://www.taobao.com/list/item/dS9Jd2NlK1hTTmNtb2VUQ1g2OUptZz09.htm 

Software Design:

ROS2 humble motor control(robot controlling):
https://control.ros.org/humble/doc/getting_started/getting_started.html
https://control.ros.org/rolling/index.html
https://docs.ros.org/en/humble/index.html

NanoOwl(Object recognition):
https://github.com/NVIDIA-AI-IOT/ROS2-NanoOWL

Lidar(distance measurement):
https://wiki.ros.org/rplidar
https://docs.ros.org/en/humble/p/rplidar_ros/

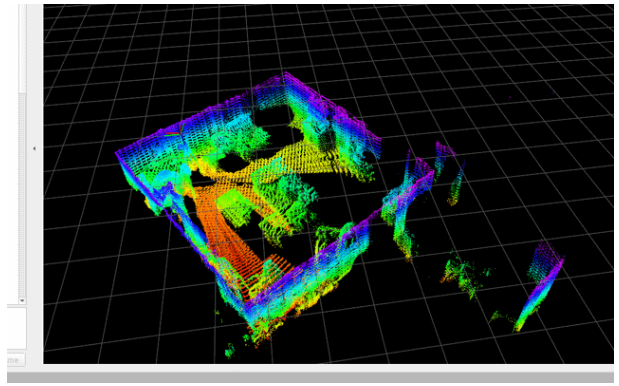
Rviz(navigation and sensing):
https://docs.ros.org/en/humble/Tutorials/Intermediate/RViz/RViz-Main.html
Gazebo(3D movement simulation)
https://docs.ros.org/en/humble/Tutorials/Advanced/Simulators/Gazebo/Simulation-Gazebo.html
https://control.ros.org/humble/doc/gazebo_ros2_control/doc/index.html
Nav2(navigation framework):
https://docs.nav2.org/

Movelt 2 (motion planning framework):
https://moveit.picknik.ai/humble/index.html

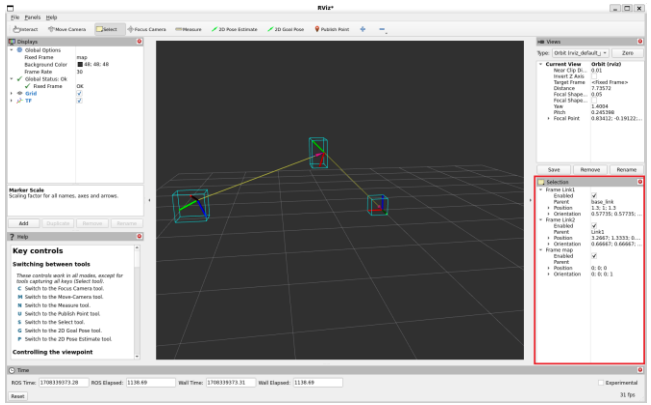
Motor



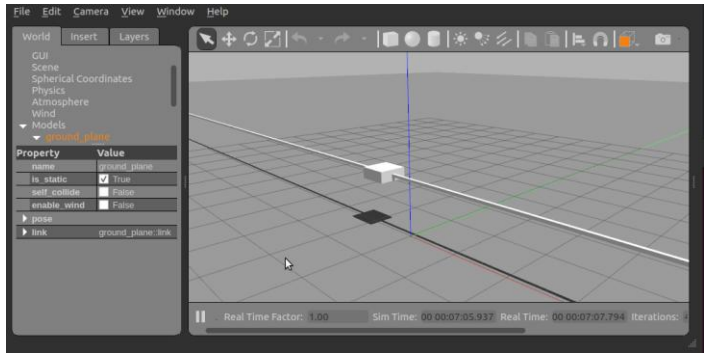
Lidar



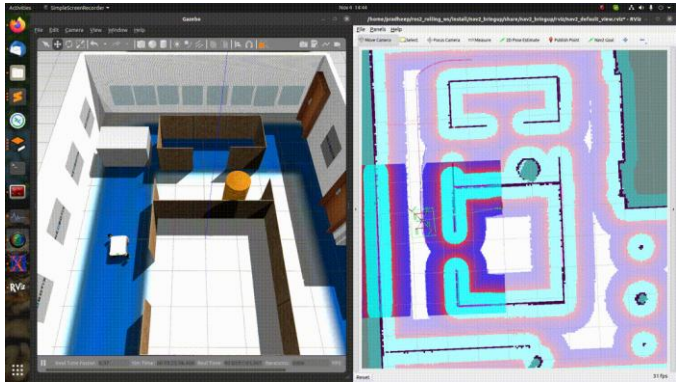
Rviz



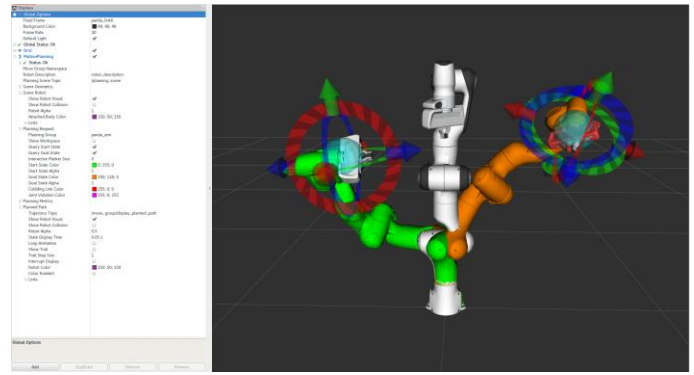
Gazebo



Nav2



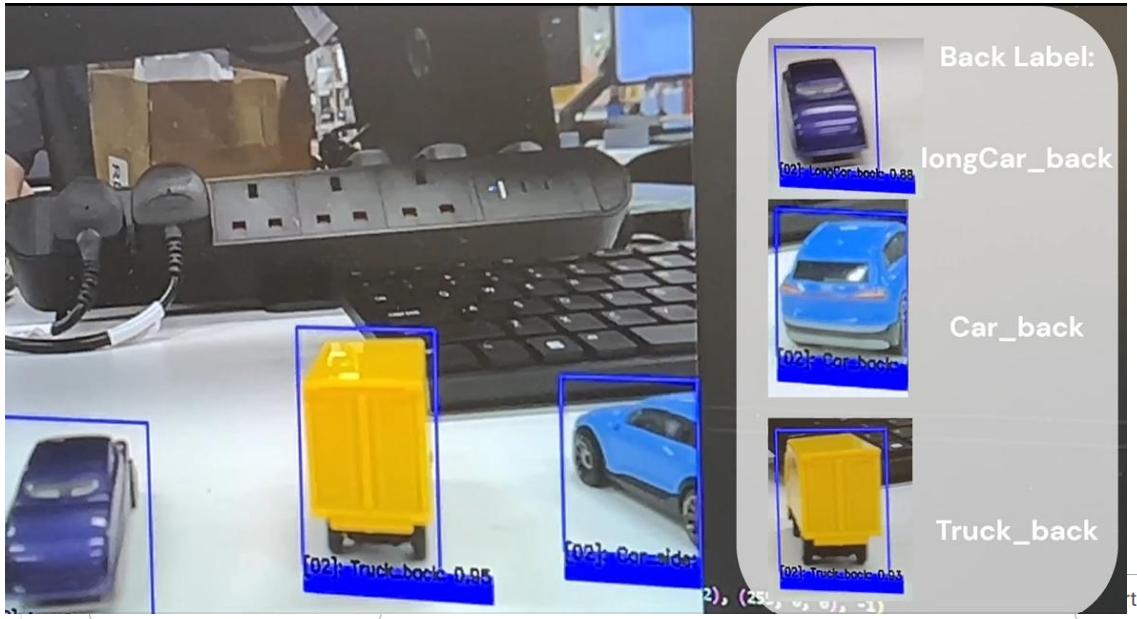
Movelt 2



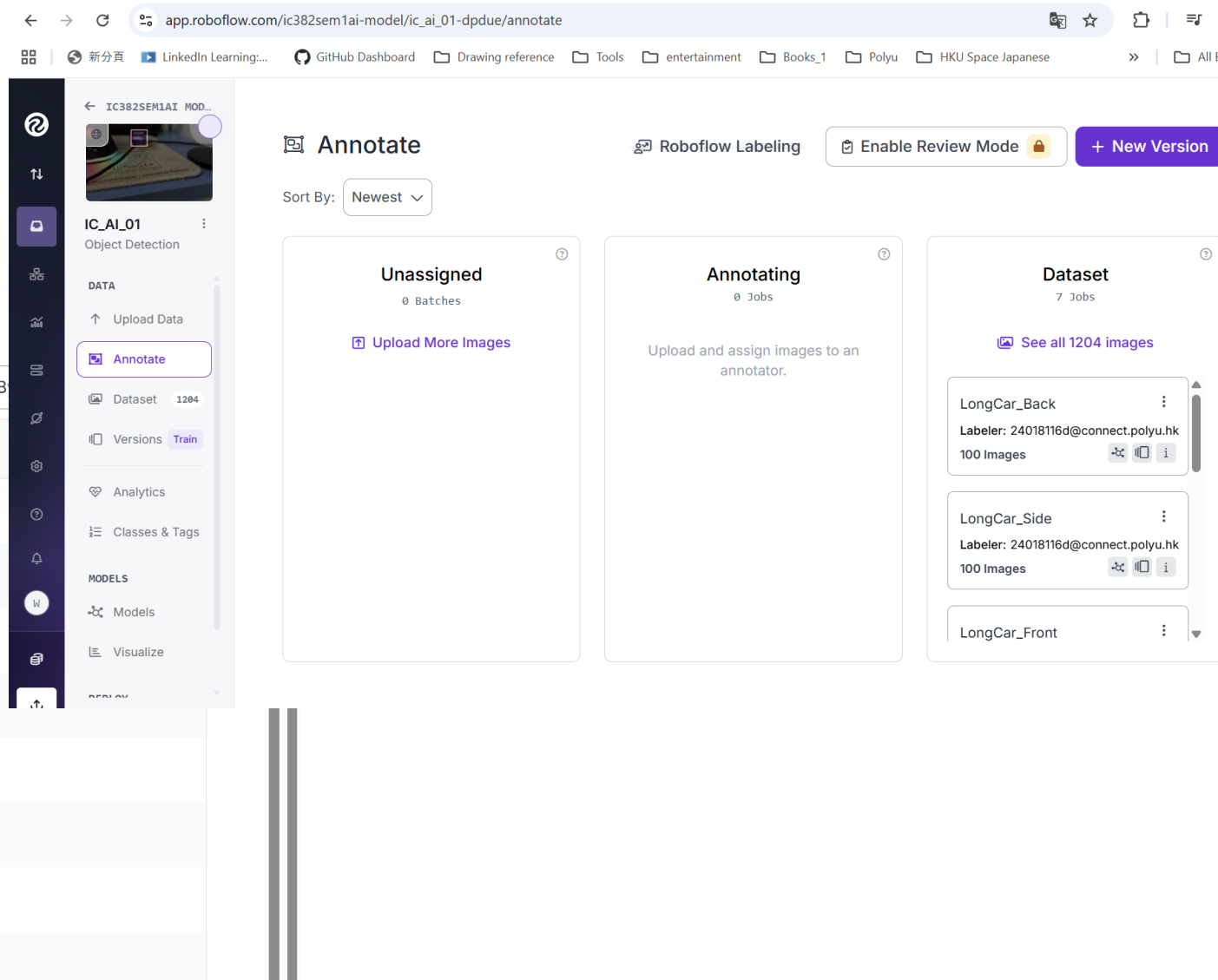
Resources for ROS2 supported tools

NAV2: https://docs.nav2.org/index.html
Moveit2: https://moveit.picknik.ai/main/index.html
ROS2 Humble: https://docs.ros.org/en/humble/index.html
URDF :https://docs.ros.org/en/humble/Tutorials/Intermediate/URDF/URDF-Main.html
TF2:https://docs.ros.org/en/humble/Tutorials/Intermediate/Tf2/Tf2-Main.html
Gazebo Fortress:https://gazebosim.org/docs/fortress/tutorials/
Rviz: https://docs.ros.org/en/humble/Tutorials/Intermediate/RViz/RViz-Main.html

AI model training



COLOR	CLASS NAME	COUNT
	Car_back	100
	Car_front	100
	Car_side	100
	LongCar_back	200
	longcar_front	200
	LongCar_Side	200
	Truck_back	104
	Truck_front	100



Defined labels for testing

3 object:

-longCar

-truck

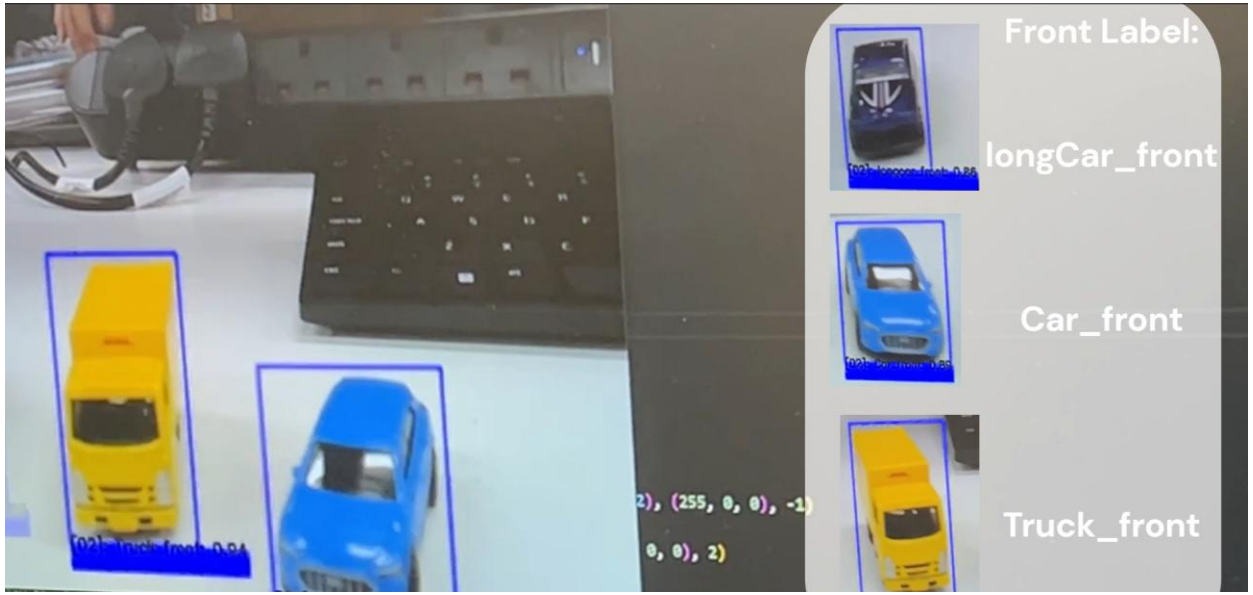
-car

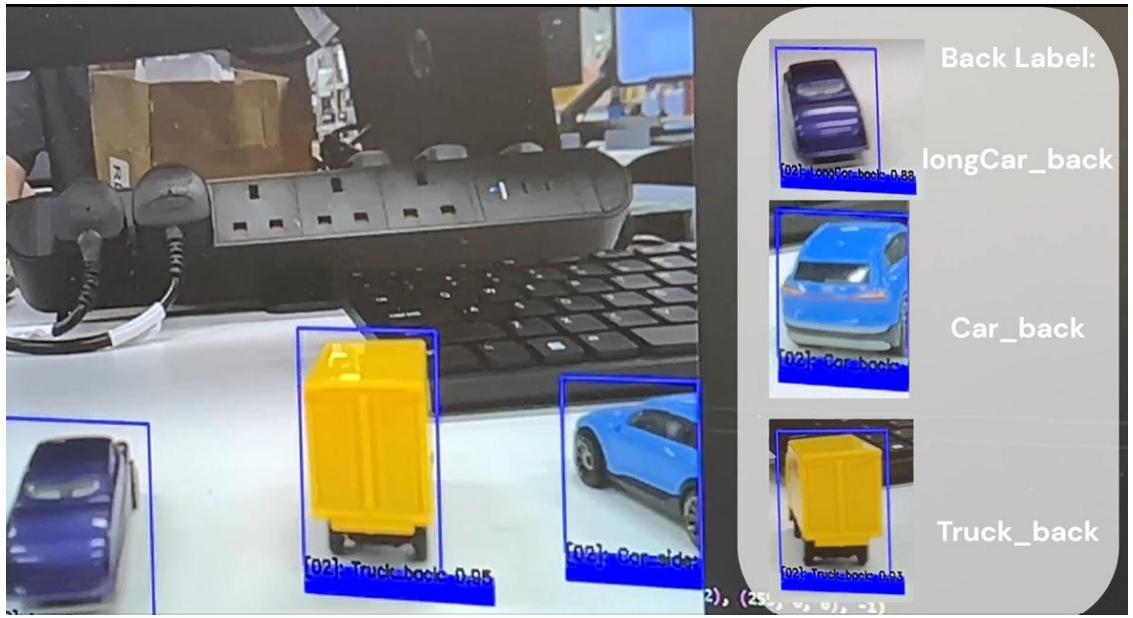
3 views:

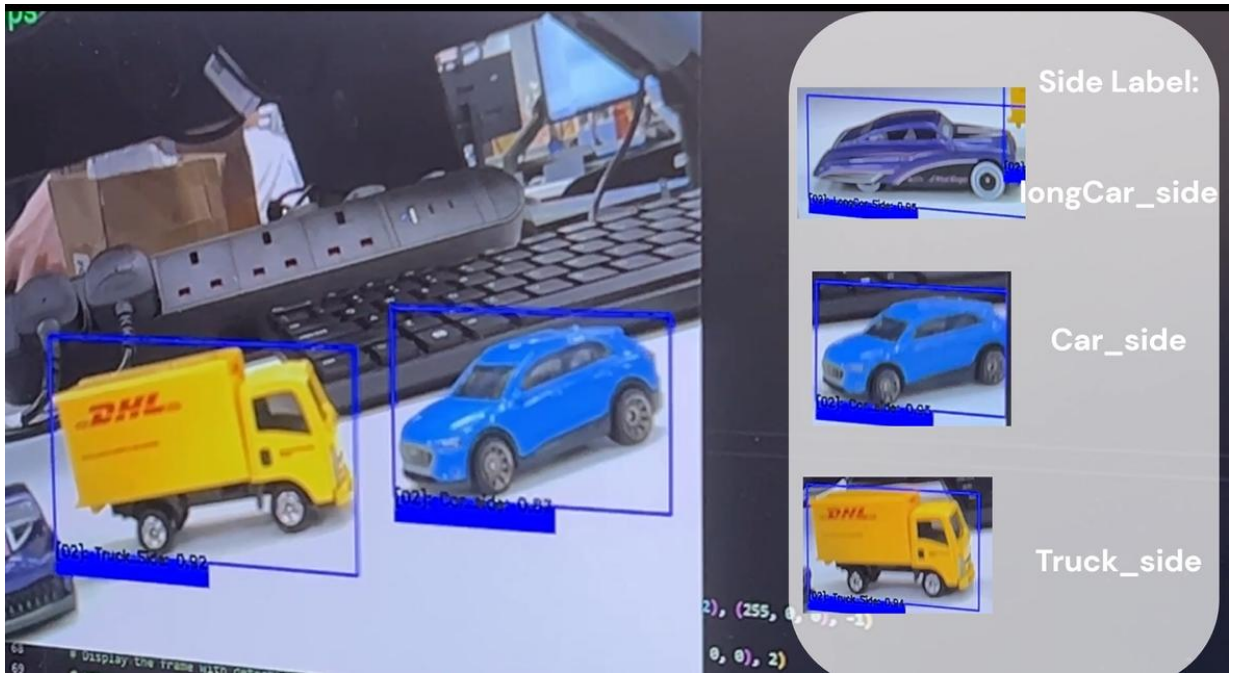
-front

-back

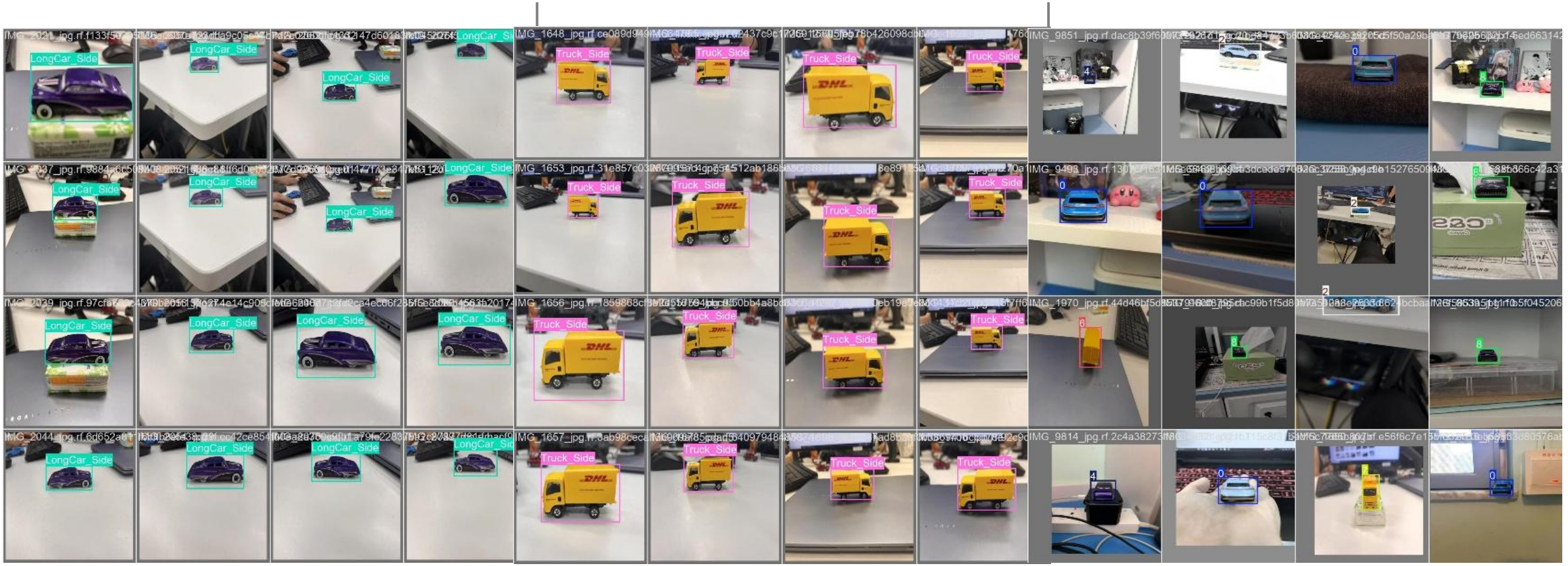
-side(left and right side view)



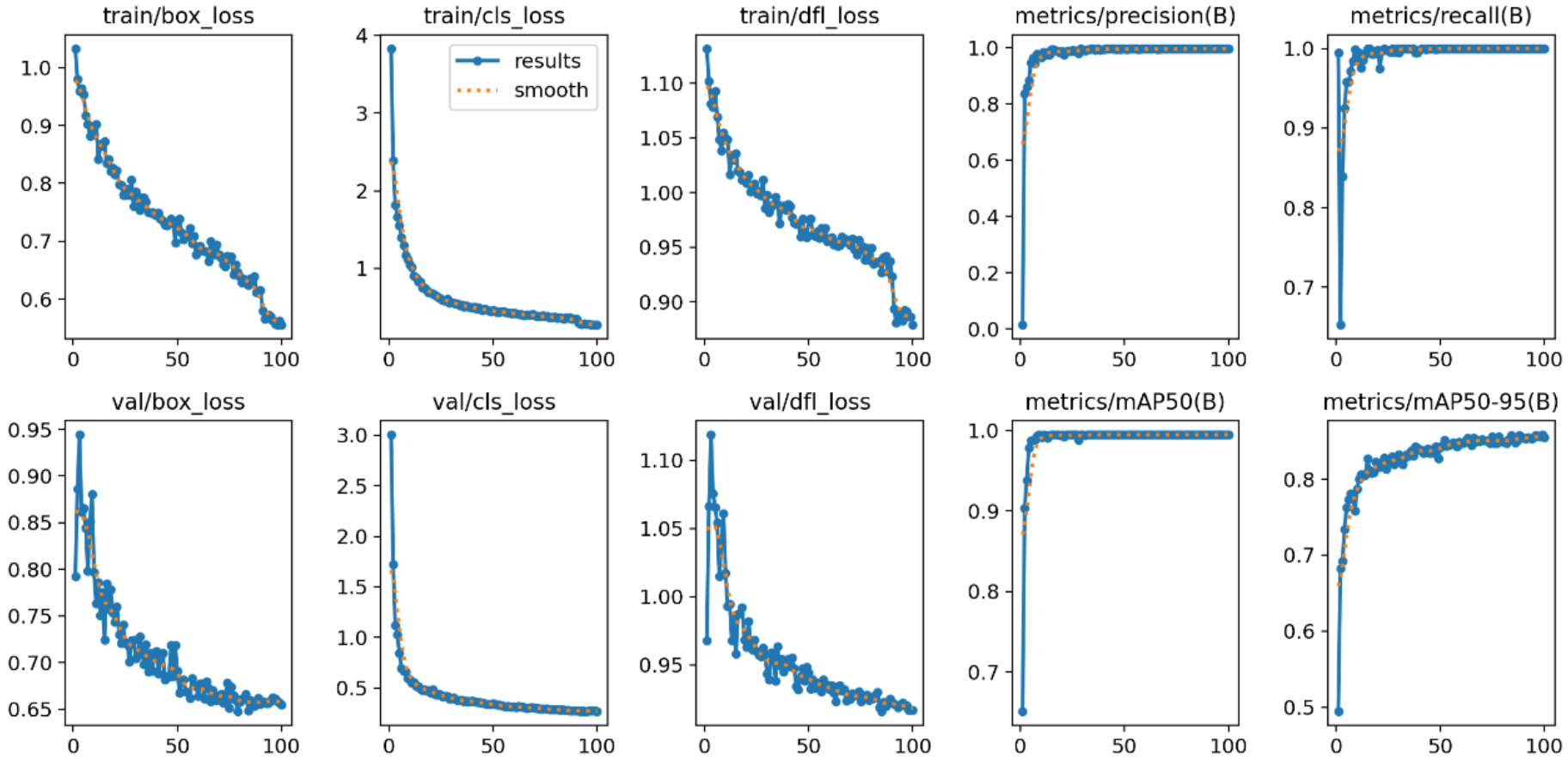




Object Detection: Toy Car



Train result (ver2)(1200 image, 9 labels, 3type of toy cars, 3 views, around 300-400 image for each car):



AI model summary:

Confidence: 0.8 -0.9

mAP50: above 80%

Scenario: varying distances are also detectable

